

Política determinista

$$\pi: S \rightarrow A$$

$$\pi_1(s_0) = \text{quit}$$

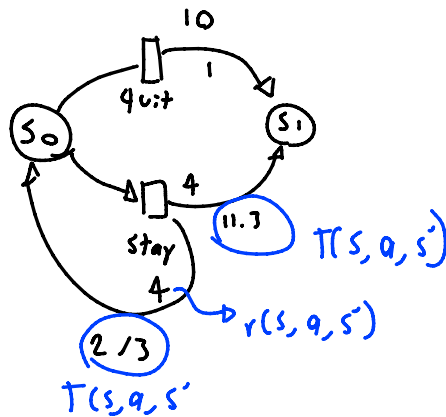
$$\pi_2(s_0) = \text{stay}$$

$$E^{\pi_1}(R_t) = 10$$

$$E^{\pi_2}(R_t) = \frac{1}{3} \cdot 4 +$$

$$\frac{2}{3} \left[\frac{1}{3} \cdot 4 + \frac{2}{3} \right]$$

$$\left[\frac{1}{3} \cdot 4 + \frac{2}{3} \right]$$



$$\frac{1}{3} \cdot 4 + \frac{2}{3} \cdot \frac{1}{3} \cdot 4 + \frac{2}{3} \cdot \frac{2}{3} \cdot \frac{1}{3} \cdot 4 + \frac{2}{3} \cdot \frac{2}{3} \cdot \frac{2}{3} \cdot \frac{1}{3} \cdot 4 + \dots$$

$$\frac{1}{3} \sum_{i=0}^{\infty} \left(\frac{2}{3}\right)^i \cdot 4 = \frac{4}{3} \sum_{i=0}^{\infty} \left(\frac{2}{3}\right)^i = \frac{4}{3} + \frac{4}{3} \sum_{i=1}^{\infty} \left(\frac{2}{3}\right)^i$$

0

$$\frac{1}{3} \cdot 4 +$$

$$\frac{2}{3} \cdot \frac{1}{3} \cdot 8 +$$

$$\frac{1}{3} \cdot 4 + \frac{2}{9} \cdot 8 + \frac{4}{27} \cdot 12 + \frac{6}{81} \cdot 16 + \frac{8}{243} \cdot 20 + \dots = 12$$

$$\frac{2}{3} \cdot \frac{2}{3} \cdot \frac{1}{3} \cdot 12 +$$

$$\frac{2}{3} \cdot \frac{2}{3} \cdot \frac{2}{3} \cdot \frac{1}{3} \cdot 16 +$$

$$\frac{2}{3} \cdot \frac{2}{3} \cdot \frac{2}{3} \cdot \frac{2}{3} \cdot \frac{1}{3} \cdot 20 +$$

$$E[R_t]$$

γ : variable de descuento $0 \leq \gamma \leq 1$ (default 1)

$$R_t = \gamma^0 r_t + \gamma^1 r_{t+1} + \gamma^2 r_{t+2} + \dots$$

Funcion de valores de estado

$$V^{\pi}(s) = E^{\pi}(R_t | S_t = s)$$

$$\pi_1 \prec \pi_2$$

si y solo si:

$$V^{\pi_1}(s) \leq V^{\pi_2}(s) \forall s \in S$$

Una política π^* es óptima si

$$V^{\pi}(s) \leq V^{\pi^*}(s) \quad \forall \pi$$